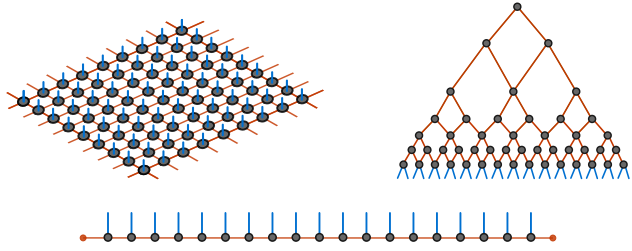
Classical compression of many-body quantum states

with tensor networks

Antoine Tilloy

Max Planck Institute of Quantum Optics, Garching, Germany



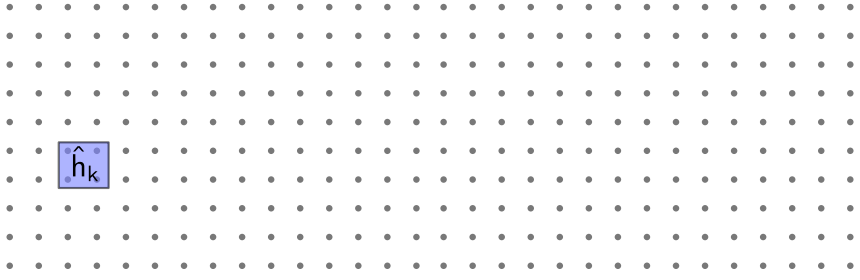
Séminaire de physique théorique

CEA Grenoble

June 2nd, 2021

MPQ

Many-body problem



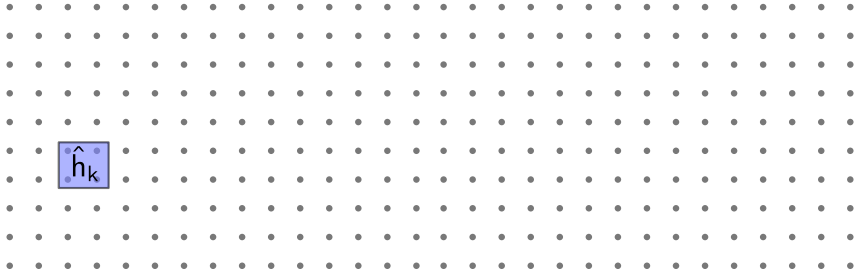
Problem

Finding low energy states of

$$\hat{H} = \sum_{k=1}^{N} \hat{h}_k$$

is **hard** because $\dim \mathscr{H} \propto d^N$

Many-body problem



Problem

Finding low energy states of

$$\hat{H} = \sum_{k=1}^{N} \hat{h}_k$$

is **hard** because $\dim \mathscr{H} \propto d^N$

Possible solutions

- Perturbation theory
- Monte Carlo
- Bootstrap IR fixed point
- **Variational optimization** (*e.g.* Mean Field, TCSA, TN)

Hard instances of the quantum many-body problem

in Nature

1 – **Condensed matter physics**

- High T_c superconductivity
- Fractional quantum Hall effect

Hard instances of the quantum many-body problem

in Nature

1 – **Condensed matter physics**

- High T_c superconductivity
- Fractional quantum Hall effect

2 – **High energy physics**

- Quantum Chromodynamics (QCD)
- "low energy" nuclear physics

Hard instances of the quantum many-body problem

in Nature

1 – **Condensed matter physics**

- High T_c superconductivity
- Fractional quantum Hall effect

2 – **High energy physics**

- Quantum Chromodynamics (QCD)
- "low energy" nuclear physics

but also quantum chemistry, interacting quantum gases...
and also voluntarily hard instances (quantum simulators, quantum computers)

1 slide summary

Many instances of the many-body problem are solvable classically

1. The quantum many-body problem can be solved with the variational method
2. The variational method is efficient if the Hilbert space can be compressed
3. Interesting quantum states in cond. matt. not generic $\Longrightarrow$ compressible
4. Tensor network states achieve the expected compression
5. This strategy can be extended to high-energy physics

1 slide summary

Many instances of the many-body problem are solvable classically

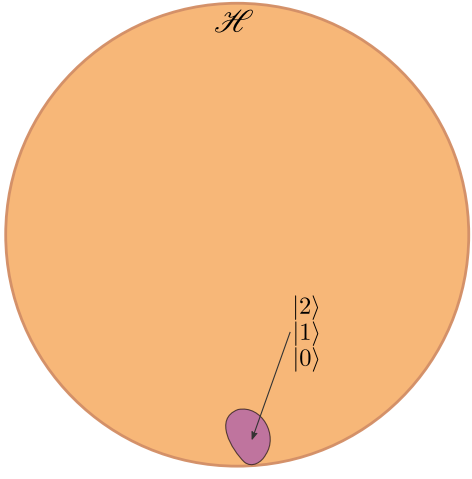
1. The quantum many-body problem can be solved with the variational method
2. The variational method is efficient if the Hilbert space can be compressed
3. Interesting quantum states in cond. matt. not generic $\implies$ compressible
4. Tensor network states achieve the expected compression
5. This strategy can be extended to high-energy physics

Take home for busy people

Exponentially large state space $\not\Rightarrow$ *hard* if *compression possible*

The variational method

Variational optimization



Generic state $\in \mathscr{H} = (\mathbb{C}^d)^{\otimes N}$:

$$|\psi\rangle = \sum_{i_1, i_2, \cdots, i_n} c_{i_1, i_2, \cdots, i_N} \, |i_1, \cdots, i_N\rangle$$

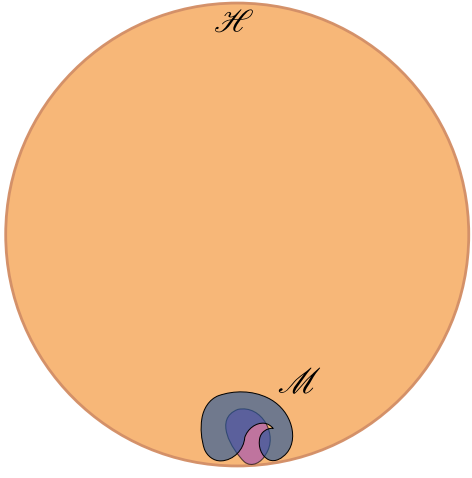
Exact variational optimization

To find the ground state:

$$|0\rangle = \min_{|\psi\rangle \in \mathscr{H}} \frac{\langle\psi|H|\psi\rangle}{\langle\psi|\psi\rangle}$$

- $\dim \mathscr{H} = d^N$

Variational optimization



Generic state $\in \mathscr{H} = (\mathbb{C}^d)^{\otimes N}$:

$$|\psi\rangle = \sum_{i_1, i_2, \cdots, i_n} c_{i_1, i_2, \cdots, i_N} \, |i_1, \cdots, i_N\rangle$$

Approx. variational optimization

To find the ground state:

$$|0\rangle = \min_{|\psi\rangle \in \mathscr{M}} \frac{\langle\psi|H|\psi\rangle}{\langle\psi|\psi\rangle}$$

- $\dim \mathscr{M} \propto \mathrm{Poly}(N)$ or fixed

An idea popular in many fields

- **Mean field** approximation (of which TNS are an extension)

$$\psi(x_1, x_2, \cdots, x_n) = \psi_1(x_1)\,\psi_2(x_2)\cdots\psi_n(x_n)$$

An idea popular in many fields

- **Mean field** approximation (of which TNS are an extension)

$$\psi(x_1, x_2, \cdots, x_n) = \psi_1(x_1)\,\psi_2(x_2)\cdots\psi_n(x_n)$$

- Special variational wave functions in **Quantum chemistry** (whole industry of ansatz)

An idea popular in many fields

- **Mean field** approximation (of which TNS are an extension)

$$\psi(x_1, x_2, \cdots, x_n) = \psi_1(x_1)\,\psi_2(x_2)\cdots\psi_n(x_n)$$

- Special variational wave functions in **Quantum chemistry** (whole industry of ansatz)
- **Moore-Read wavefunctions** in the study of the quantum Hall effect

$$\psi(x_1, x_2, \cdots, x_n) = \left\langle \hat{\phi}(x_1)\hat{\phi}(x_2)\cdots\hat{\phi}(x_n) \right\rangle_{\mathrm{CFT}}$$

An idea popular in many fields

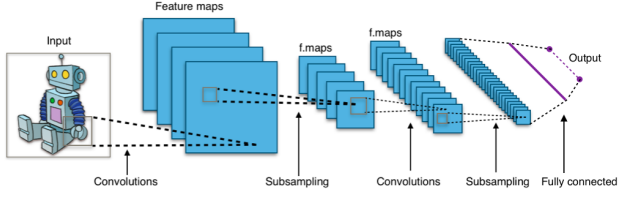
- **Mean field** approximation (of which TNS are an extension)

$$\psi(x_1, x_2, \cdots, x_n) = \psi_1(x_1)\,\psi_2(x_2)\cdots\psi_n(x_n)$$

- Special variational wave functions in **Quantum chemistry** (whole industry of ansatz)
- **Moore-Read wavefunctions** in the study of the quantum Hall effect

$$\psi(x_1, x_2, \cdots, x_n) = \left\langle \hat{\phi}(x_1)\hat{\phi}(x_2)\cdots\hat{\phi}(x_n) \right\rangle_{\mathrm{CFT}}$$

- Fully connected and convolutional **neural networks** used in machine learning



State compression and the area law

The room for compression

atypicality $\Longrightarrow$ compressibility

The room for compression

atypicality $\Longrightarrow$ compressibility

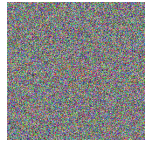
- For image classification, exponentially many classifiers $N_{\text{colors}}^{N_{\text{pixels}}}$ but



cat image



dog image



“typical” image

The room for compression

atypicality $\Longrightarrow$ compressibility

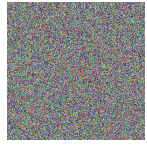
- For image classification, exponentially many classifiers $N_{\text{colors}}^{N_{\text{pixels}}}$ but



cat image

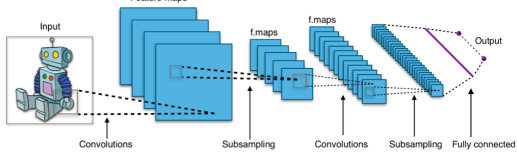


dog image



"typical" image

- efficient classifiers $f(x)$ with only $Poly(N_{\text{pixels}}) \ll N_{\text{colors}}^{N_{\text{pixels}}}$ parameters



The room for compression

atypicality $\Longrightarrow$ compressibility

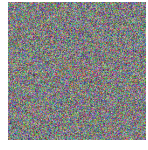
- For image classification, exponentially many classifiers $N_{\text{colors}}^{N_{\text{pixels}}}$ but



cat image

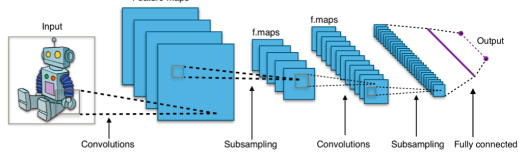


dog image



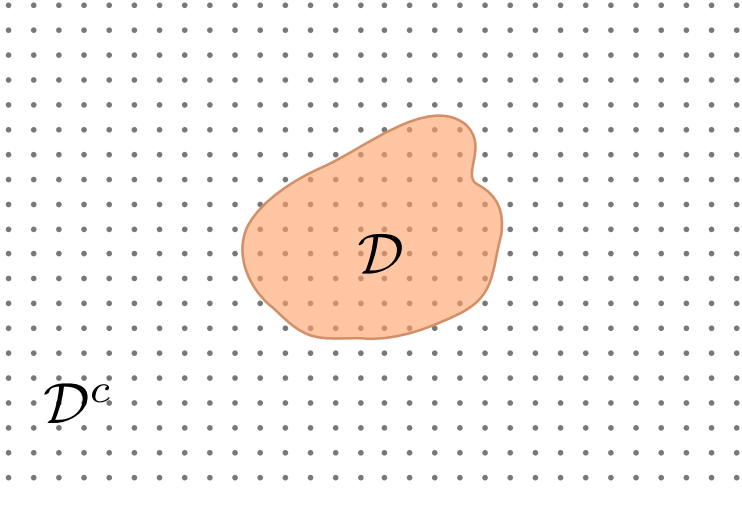
"typical" image

- efficient classifiers $f(x)$ with only $Poly(N_{\text{pixels}}) \ll N_{\text{colors}}^{N_{\text{pixels}}}$ parameters



- What is the atypicality analog for quantum? $\rightarrow$ **Entanglement**

Interesting states are weakly entangled



Low energy state

$|\psi\rangle = |0\rangle$ or $|1\rangle$ ...

Reduced density matrix

$\rho = \mathrm{tr}_{\mathcal{D}^c}\left[|\psi\rangle\langle\psi|\right]$

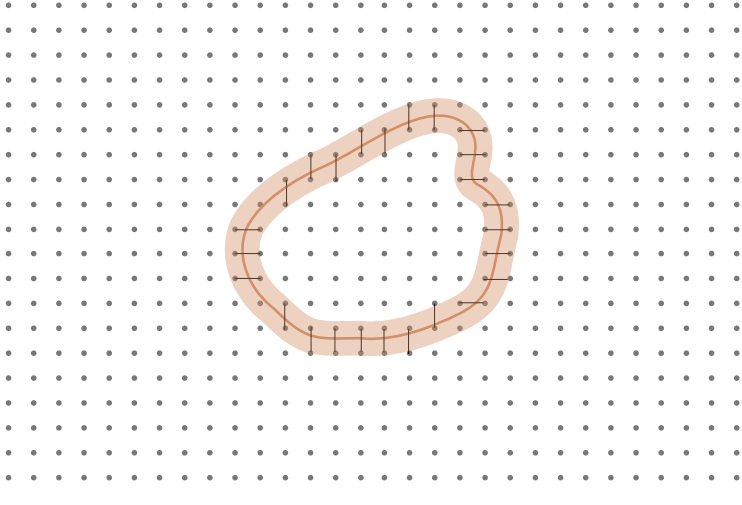
Entanglement entropy

$S = -\mathrm{tr}\left[\rho \log \rho\right]$

Area law

$S \propto |\partial\mathcal{D}|$

Interesting states are weakly entangled



Low energy state

$|\psi\rangle = |0\rangle$ or $|1\rangle$ ...

Reduced density matrix

$\rho = \mathrm{tr}_{\mathcal{D}^c}\left[|\psi\rangle\langle\psi|\right]$

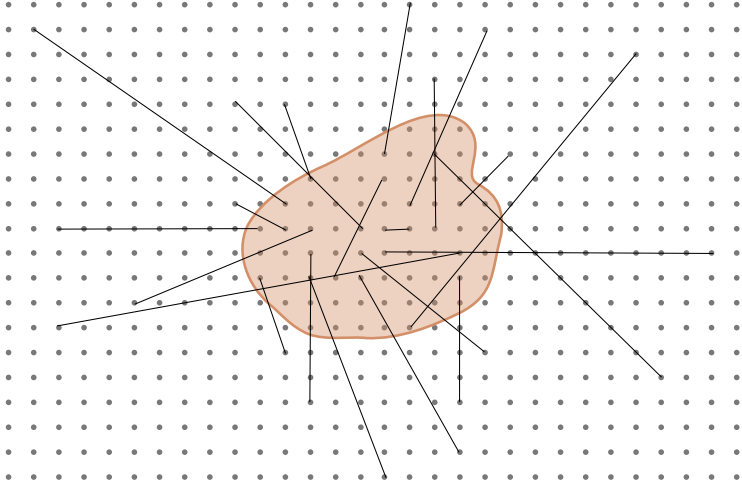
Entanglement entropy

$S = -\mathrm{tr}\left[\rho \log \rho\right]$

Area law

$S \propto |\partial\mathcal{D}|$

Typical states are strongly entangled



Random state

$|\psi\rangle = U_{\text{Haar}}|\text{trivial}\rangle$

Reduced density matrix

$\rho = \text{tr}_{\mathcal{D}^c}\left[|\psi\rangle\langle\psi|\right]$

Entanglement entropy

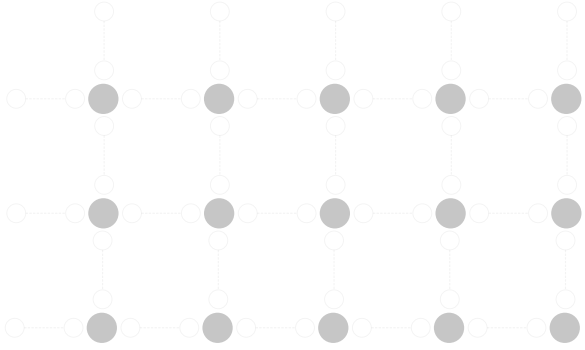
$S = -\text{tr}\left[\rho \log \rho\right]$

Volume law

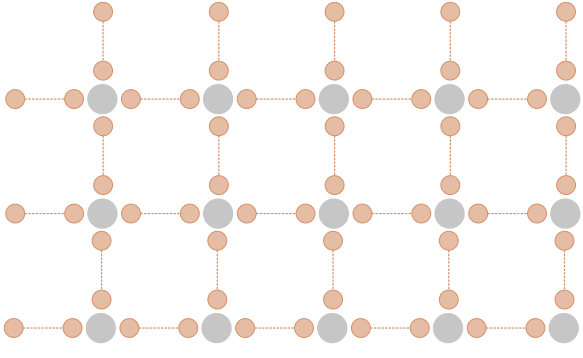
$S \propto |\mathcal{D}|$

Tensor network states

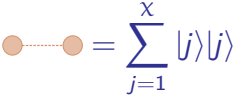
Constructing weakly entangled states



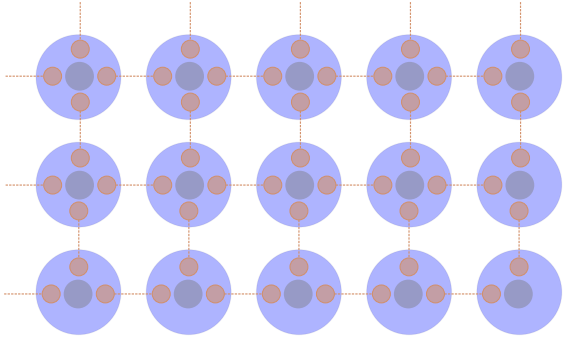
Constructing weakly entangled states



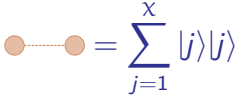
1. Put auxiliary **maximally entangled** states between sites



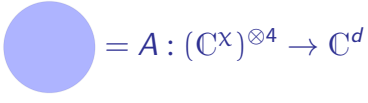
Constructing weakly entangled states



1. Put auxiliary **maximally entangled** states between sites

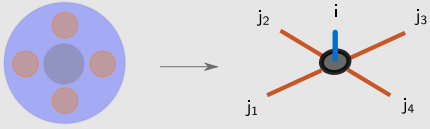


2. Map to initial Hilbert space on each site



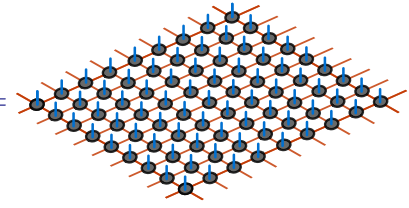
Tensor network states: definition

Why "tensor" network?



$A : (\mathbb{C}^X)^{\otimes 4} \to \mathbb{C}^d \quad \longrightarrow \quad A^i_{j_1,j_2,j_3,j_4}$

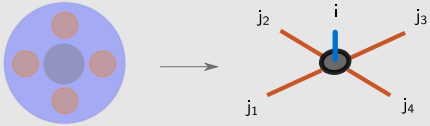
$|A\rangle =$



with tensor contractions on links

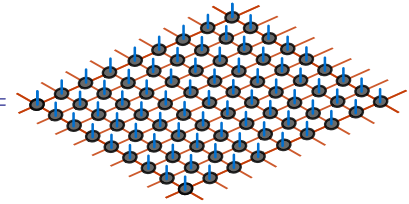
Tensor network states: definition

Why "tensor" network?



$A : (\mathbb{C}^\chi)^{\otimes 4} \to \mathbb{C}^d \quad \longrightarrow \quad A^i_{j_1,j_2,j_3,j_4}$

$|A\rangle =$



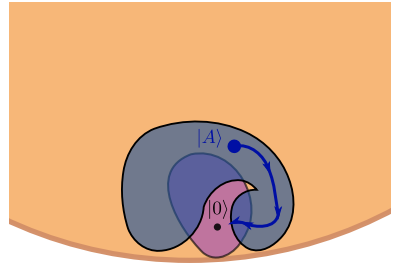
with tensor contractions on links

Optimization

Find best A for fixed χ $\quad (d \times \chi^4$ coeff.)

$$E_0 \simeq \min_A \frac{\langle A|\hat{H}|A\rangle}{\langle A|A\rangle}$$

for example go down $\frac{\partial E}{\partial A^i_{j_1,j_2,j_3,j_4}}$

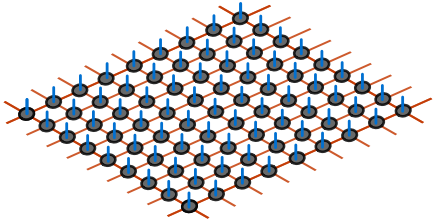


Generalizations: different tensor networks

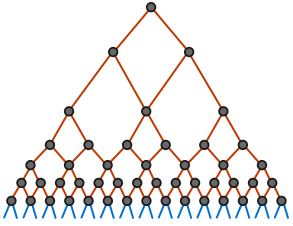
Matrix Product States (MPS)



Projected Entangled Pair States (PEPS)

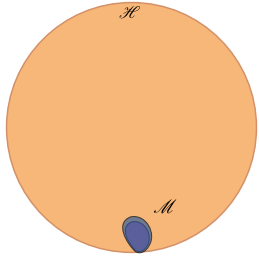


Multi-scale Entanglement Renormalization Ansatz (MERA)



Some facts

$d = 1$ spatial dimension

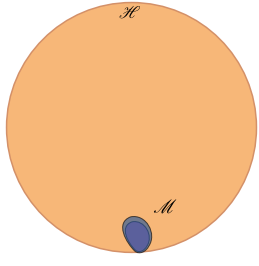


Theorems (colloquially)

1. For gapped H, TNS $|A\rangle$ approximate well $|0\rangle$ with χ fixed
2. **All** $|A\rangle$ are ground states of gapped H

Some facts

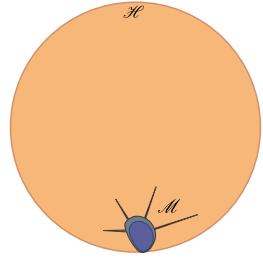
$d = 1$ spatial dimension



Theorems (colloquially)

1. For gapped H, TNS $|A\rangle$ approximate well $|0\rangle$ with χ fixed
2. **All** $|A\rangle$ are ground states of gapped H

$d \geqslant 2$ spatial dimension



Folklore

1. For gapped H, TNS $|A\rangle$ approximate well $|0\rangle$ with χ fixed
2. **Most** $|A\rangle$ are ground states of gapped H

State of the art

Dense: all states approximable (trivial)
Efficient: cost $\mathit{Poly}(\chi)$ error $\mathit{superPoly}(\chi^{-1})$

State of the art

Dense: all states approximable (trivial)
Efficient: cost $Poly(\chi)$ error $superPoly(\chi^{-1})$

It's all a matter of prefactors and exponents

- 1 space dimension $\rightarrow \chi \geqslant 1000 \rightarrow$ machine precision (MPS results "numerically exact")

State of the art

Dense: all states approximable (trivial)
Efficient: cost $Poly(\chi)$ error $superPoly(\chi^{-1})$

It's all a matter of prefactors and exponents

- 1 space dimension $\rightarrow \chi \geqslant 1000 \rightarrow$ machine precision
  (MPS results "numerically exact")
- 2 space dimensions $\rightarrow \chi \sim 10 \rightarrow$ efficient
  (PEPS efficient to $10^{-2} - 10^{-6}$ depending on problems)
- 3 space dimensions $\rightarrow \chi \sim 3 \rightarrow$ theoretically efficient but too expensive

State of the art

Dense: all states approximable (trivial)
Efficient: cost $Poly(\chi)$ error $superPoly(\chi^{-1})$

It's all a matter of prefactors and exponents

- 1 space dimension $\rightarrow \chi \geqslant 1000 \rightarrow$ machine precision
  (MPS results "numerically exact")
- 2 space dimensions $\rightarrow \chi \sim 10 \rightarrow$ efficient
  (PEPS efficient to $10^{-2} - 10^{-6}$ depending on problems)
- 3 space dimensions $\rightarrow \chi \sim 3 \rightarrow$ theoretically efficient but too expensive

State of the art

Dense: all states approximable (trivial)
Efficient: cost $Poly(\chi)$ error $superPoly(\chi^{-1})$

It's all a matter of prefactors and exponents

- 1 space dimension $\rightarrow \chi \geqslant 1000 \rightarrow$ machine precision
  (MPS results "numerically exact")
- 2 space dimensions $\rightarrow \chi \sim 10 \rightarrow$ efficient
  (PEPS efficient to $10^{-2} - 10^{-6}$ depending on problems)
- 3 space dimensions $\rightarrow \chi \sim 3 \rightarrow$ theoretically efficient but too expensive

Beyond ground states

- Low-lying spectrum - excited states (on the G.S. tangent space)
- Thermal states (because area law)
- Real-time evolution (but no long time quench)

Beyond area law (no free lunch)

A sad fact

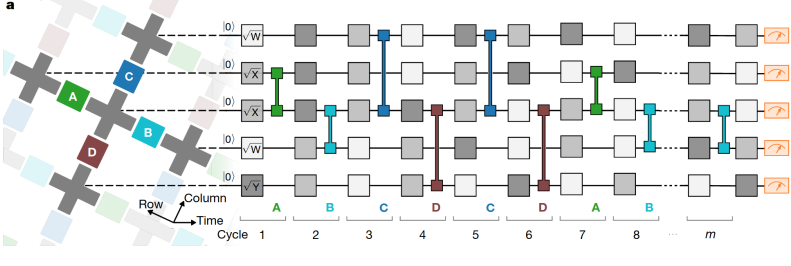
Scrambling of real-time evolution creates volume law entanglement even with local gapped H

Beyond area law (no free lunch)

A sad fact

Scrambling of real-time evolution creates volume law entanglement even with local gapped H

Motivates Google supremacy experiment [Nature, 2020]

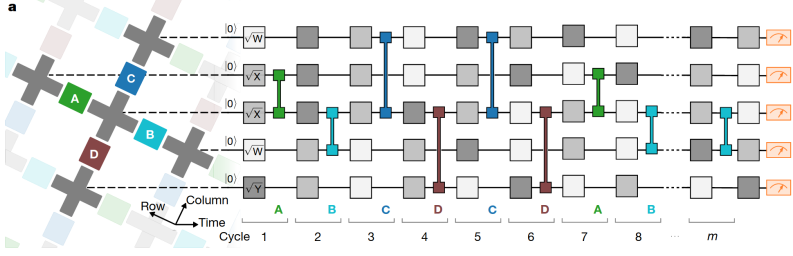


Beyond area law (no free lunch)

A sad fact

Scrambling of real-time evolution creates volume law entanglement even with local gapped H

Motivates Google supremacy experiment [Nature, 2020]



Still possible compression [Zhou, Stoudenmire, Waintal, PRX 2020]
but no exponential miracle **IF** errors low enough

Impact for quantum technologies

Applications of quantum computers/simulators to condensed matter problems often mentioned $\rightarrow$ "solve 2d-Fermi-Hubbard"

Impact for quantum technologies

Applications of quantum computers/simulators to condensed matter problems often mentioned $\rightarrow$ "solve 2d-Fermi-Hubbard"

IF in area law phase (e.g. ground state, thermal states, low lying spectrum, short real-time evolution for local H)

- Quantum good as comparison benchmark
- Probably not asymptotically competitive with classical

Impact for quantum technologies

Applications of quantum computers/simulators to condensed matter problems often mentioned $\rightarrow$ "solve 2d-Fermi-Hubbard"

IF in area law phase (e.g. ground state, thermal states, low lying spectrum, short real-time evolution for local H)

- Quantum good as comparison benchmark
- Probably not asymptotically competitive with classical

Problems likely to remain classically hard in practice

- Quantum quenches (because real-time)
- Factoring (same, error corrected Shor creates volume law)
- Quantum chemistry (because H not local)

Impact for quantum technologies

Applications of quantum computers/simulators to condensed matter problems often mentioned $\rightarrow$ "solve 2d-Fermi-Hubbard"

IF in area law phase (e.g. ground state, thermal states, low lying spectrum, short real-time evolution for local H)

- Quantum good as comparison benchmark
- Probably not asymptotically competitive with classical

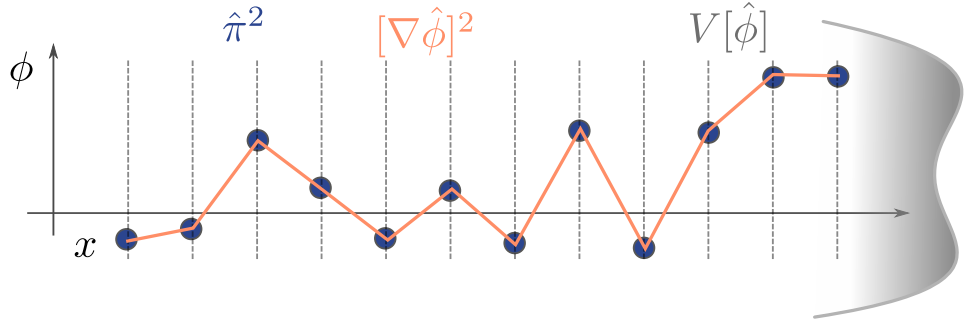
Problems likely to remain classically hard in practice

- Quantum quenches (because real-time)
- Factoring (same, error corrected Shor creates volume law)
- Quantum chemistry (because H not local)

what about quantum field theory?

Quantum field theory

Intuitive definition: canonical quantization



Hamiltonian

A continuum of nearest neighbor coupled anharmonic oscillators

$$\hat{H} = \int_{\mathbb{R}^d} \mathrm{d}^d x \quad \underbrace{\frac{\hat{\pi}(x)^2}{2}}_{\text{on-site inertia}} + \underbrace{\frac{[\nabla\hat{\phi}(x)]^2}{2}}_{\text{spatial stiffness}} + \underbrace{V(\hat{\phi}(x))}_{\text{on-site potential}}$$

with canonical commutation relations $[\hat{\phi}(x), \hat{\pi}(y)] = i\delta^d(x-y)\mathbb{1}$ (*i.e.* bosons)

Quantum field theory in a nutshell

QFT

A relativistic QFT is just a condensed matter problem on an infinitely fine grained lattice with peculiarities

1. H such that the dynamics is Lorentz covariant
2. Critical at short distances (scale invariant when zooming in)
3. Divergent when naively defined

e.g. QCD is a strongly correlated (infinitely) many-body problem.

Two strategies

Numerical continuum limit

1. Discretize the QFT on a fine lattice (spacing ε)
2. Solve by compressing the state with standard TNS
3. Extrapolate results $\varepsilon \to 0$

(same strategy as lattice MC)

Two strategies

Numerical continuum limit

1. Discretize the QFT on a fine lattice (spacing ε)
2. Solve by compressing the state with standard TNS
3. Extrapolate results $\varepsilon \to 0$

(same strategy as lattice MC)

Analytical continuum limit

1. Take the continuum limit of TNS
2. Use the variational method on the true continuum Hamiltonian

more ambitious strategy

Two strategies

Numerical continuum limit

1. Discretize the QFT on a fine lattice (spacing ε)
2. Solve by compressing the state with standard TNS
3. Extrapolate results $\varepsilon \to 0$

(same strategy as lattice MC)

Analytical continuum limit

1. Take the continuum limit of TNS
2. Use the variational method on the true continuum Hamiltonian

more ambitious strategy

Fenyman was pessimistic (1986). No class of wave functions can be:

- Extensive
- Computable
- Deal with the short distance scale invariance of QFT

Analytical continuum limit – long story short

- 2010 F. Verstraete and I. Cirac *MPS* $\rightarrow$ *CMPS* in 1 space dimension

Analytical continuum limit – long story short

- 2010 F. Verstraete and I. Cirac *MPS* $\to$ *CMPS* in 1 space dimension
- 2019 AT and I. Cirac *PEPS* $\to$ *CPEPS* in arbitrary space dimension

Analytical continuum limit – long story short

- 2010 F. Verstraete and I. Cirac *MPS* $\rightarrow$ *CMPS* in 1 space dimension
- 2019 AT and I. Cirac *PEPS* $\rightarrow$ *CPEPS* in arbitrary space dimension
- 2021 AT *CMPS* $\rightarrow$ *RCMPS* deal with relativistic QFT divergences

Analytical continuum limit – long story short

- 2010 F. Verstraete and I. Cirac *MPS* $\to$ *CMPS* in 1 space dimension
- 2019 AT and I. Cirac *PEPS* $\to$ *CPEPS* in arbitrary space dimension
- 2021 AT *CMPS* $\to$ *RCMPS* deal with relativistic QFT divergences

RCMPS – relativistic continuous matrix product states

$$|R, Q\rangle = \mathrm{tr}\left\{ \mathcal{P} \exp\left[\int \mathrm{d}x \, Q \otimes \mathbb{1} + R \otimes a^\dagger(x) \right] \right\} |0\rangle_a$$

R, Q finite $D \times D$ matrices

Two papers [soon v2 with state-of-the-art numerics]

- Variational method in relativisitic QFT without cutoff (short) `arXiv:2102.07733v1`
- Relativistic continuous matrix product states for QFT without cutoff (long) `arXiv:2102.07741v1`

ϕ_2^4 testbed

Renormalized ϕ_2^4 theory

$$H = \int \mathrm{d}x \, \frac{:\pi^2:_a}{2} + \frac{:(\nabla\phi)^2:_a}{2} + \frac{m^2}{2} :\phi^2:_a + g :\phi^4:_a$$

(note that $:\diamondsuit:_a$ depends on m)

ϕ_2^4 testbed

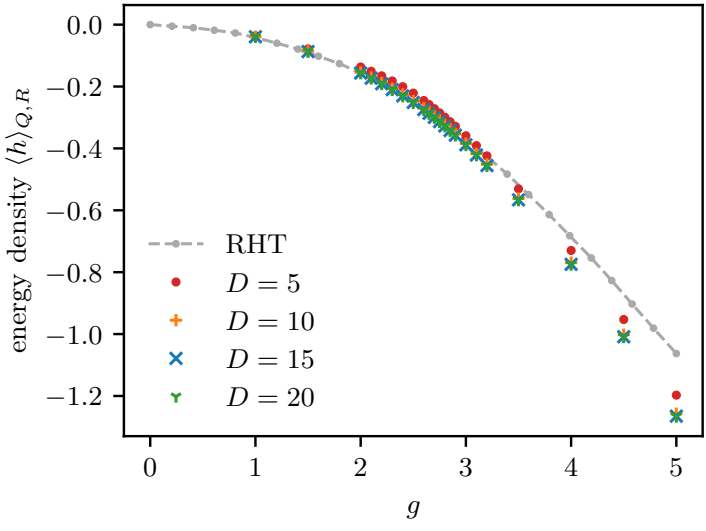
Renormalized ϕ_2^4 theory

$$H = \int \mathrm{d}x \, \frac{:\pi^2:_a}{2} + \frac{:(\nabla\phi)^2:_a}{2} + \frac{m^2}{2} :\phi^2:_a + g :\phi^4:_a$$

(note that $:\diamondsuit:_a$ depends on m)

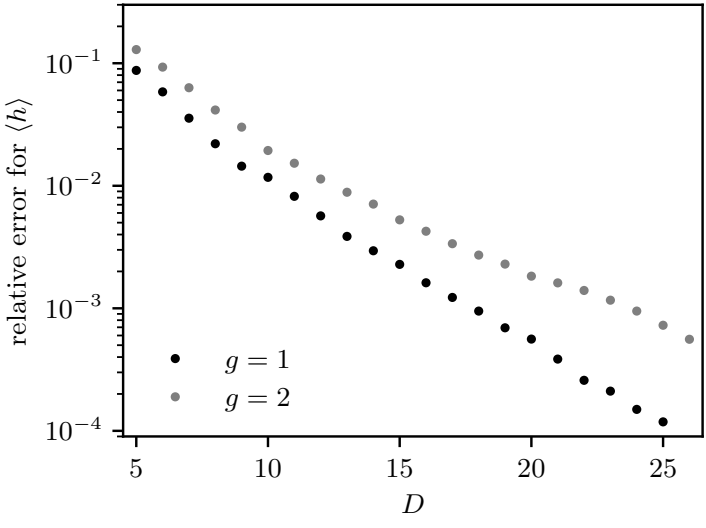
1. Rigorously defined relativistic QFT without cutoff (Wightman QFT)
2. Vacuum energy density finite
3. Very difficult to solve unless $g \ll m^2$ (perturbation theory)
4. Phase transition around $f_c = \frac{g}{4m^2} = 11$ i.e. $g \simeq 2.7$ in mass units

New results [soon to be in v2]



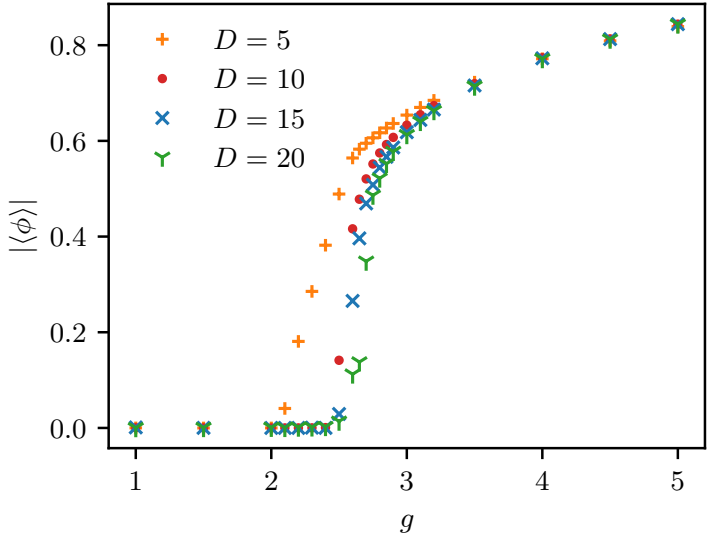
Number of parameters optimized: $2D^2$, cost $\propto D^3$

New results [soon to be in v2]



Number of parameters optimized: $2D^2$, cost $\propto D^3$

New results [soon to be in v2]



Number of parameters optimized: $2D^2$, cost $\propto D^3$

1 slide summary – revisited

Many instances of the many-body problem are solvable classically

1. Interesting (low energy) many-body quantum states verify the **area law** $S \propto |\partial\mathcal{D}|$ instead of $S \propto |\mathcal{D}|$
2. Allows to compress states into an extensive description: **tensor networks** ($2^N \to N$)
3. This can be used to classically get ground states of local H, thermal states
4. Scrambling, volume law states remain hard $\to$ Q-computers
5. Strategy usable beyond lattice to solve (so far $1+1$) relativistic QFT, bypassing Feynman no-go argument